Sujee Rubio

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EDUCATION

Northwestern University, Evanston, IL **Expected June 2027**

Bachelor of Science in Mechanical Engineering, Concentration in Aerospace

- Cumulative GPA: 3.63/4.00 | Society of Women Engineers
- **Relevant Coursework:** Orbital Mechanics, Machine Dynamics, Stress Analysis, Fluid Dynamics, Thermodynamics, Multivariable Calculus, Linear Algebra, Differential Equations, Physics (Mechanics; Electricity & Magnetism)

WORK EXPERIENCE

General Atomics Electromagnetic Systems (GA-EMS), Herndon, VA

Mechanical Engineering Intern – Satellite Dispenser Integration & Test **June 2026 – August 2026**

- Supported system-level assembly, integration, and test of a satellite dispenser that will launch and deploy CubeSats on orbit
- Verified deployment safety by running Frangibolt actuator no-fire tests at ambient and 45°C, confirming thermal-vacuum (TVAC) checks cannot trigger an accidental deployment
- Defined the payload-to-dispenser mechanical interface: designed a cup-cone interface with a hold-down and release mechanism (HDRM) within a 3×3×2 in. envelope and released GD&T drawings for fabrication
- Performed preload and joint-separation analysis against a 60g NASA ESPA design load, establishing a 450 g payload capability within 200 lbf of preload; built the test article and mass adapter for vibration testing
- Wrote the dispenser mass properties test procedure; measured center of gravity with a FARO laser tracker in hoist-based testing

Mechanical Engineering Intern – CubeSat GNC Test Systems **June 2025 – August 2025**

- Designed, built, and tested a 6′×3′ ferrofluid-based microgravity simulator used to validate CubeSat guidance, navigation, and control (GNC); developed a frictionless puck system that eliminated umbilical interference
- Ran materials tests comparing platform size, magnet strength, and ferrofluid properties to select a design with stable, frictionless motion
- Wrote test procedures for and performed center of gravity (CG) and moment of inertia (MOI) tests on mass simulators to verify spacecraft mass properties

PROJECT EXPERIENCE

High-Power Rocket Structural Analysis | *Stress Analysis Course Project* **March 2026 – June 2026**

- Verified on a 3-person team that a high-power rocket survives max-Q via coupled CFD-to-FEA analysis in ANSYS Workbench
- Wrote MATLAB scripts to process OpenRocket flight data, identifying max-Q (26.4 kPa at Mach 0.62) as the critical load case
- Ran ANSYS Static Structural FEA with CFD-imported loads, confirming a factor of safety well above 1.5

Supersonic Flow & Sonic Boom Analysis | *Independent Study, Advisor: Prof. Neelesh Patankar* **March 2026 – June 2026**

- Modeled the Concorde at Mach 2 with MATLAB oblique-shock and Prandtl-Meyer solvers, computing pressure, temperature, and Mach number nose to tail and constructing the N-wave signature (8.5 kPa bow shock, 105 ms)
- Ran parametric sweeps of nose angle vs. shock strength, finding the Mach 2 shock detachment limit at 22.9°
- Derived small-angle models from first principles and benchmarked them against exact solutions; a hybrid log-pressure/linear-Mach model cut end-state pressure error from 66% to 16%

Gearbox Shaft Design | *Theory of Machines Course Project* **March 2026 – June 2026**

- Designed the output shaft of a 70 kW two-stage spur gear reducer (2,600 → 299 rpm, 2,233 N·m) on a 3-person team, sizing the shaft, keyways, and SKF 6313 bearings via Goodman fatigue and AGMA analysis for infinite life (FOS 2.29 fatigue, 2.13 yield); created SolidWorks drawings with H7/k6 fits

LEADERSHIP

Kappa Theta Pi (Pre-Professional Technology Fraternity), *VP of DEI* **April 2024 – Present**

- Led DEI efforts during recruitment for an inclusive, well-organized process; organized women-in-tech networking sessions

Kappa Alpha Theta, *Philanthropy Director, VP of External Affairs* **January 2024 – Present**

- Created philanthropy events raising over $5,000 in two years; secured national organization and university approvals

SKILLS

Programming: MATLAB, Python

CAD & Analysis: SolidWorks, Onshape, NX Siemens, AutoCAD, ANSYS (Fluent, Mechanical), GD&T, MS Office

Test & Manufacturing: FARO laser tracker, manual/CNC mill, laser cutter, Bambu 3D printer, soldering, band saw, drill press